Lecture Notes

Lecture 10

The Spectral Theorem

Lecture 9 showed that a normal operator has orthogonal eigenvectors for distinct eigenvalues. The *spectral theorem* completes the story in the most useful way possible: a normal operator on a finite-dimensional complex space has a *complete* orthonormal basis of eigenvectors—it is diagonal in an orthonormal basis, i.e. *unitarily diagonalizable*. For observables (self-adjoint operators) the eigenvalues are moreover real. This single theorem is the mathematical backbone of the finite ideal-projective model of quantum mechanics. It supplies the eigenspace projectors. The Born probabilities and the conditional Lueders update are separate physical postulates; the general Hilbert-space theory and generalized measurements are deferred to Lecture 14.

Learning objectives

By the end of this lecture you will be able to:

- State the finite-dimensional complex and real spectral theorems and the equivalences they assert.
- Unitarily diagonalize a finite-dimensional Hermitian or complex normal operator.
- Write the finite spectral decomposition $T = \sum_k \lambda_k P_k$ and compute $f(T)$ spectrally.
- Relate commuting finite-dimensional self-adjoint observables and simultaneous diagonalization to the finite ideal-projective measurement model.

1 Schur's theorem

The proof rests on triangularizing in an *orthonormal* basis.

Theorem 10.1 (Schur). Every operator on a finite-dimensional complex inner product space has an upper-triangular matrix with respect to some orthonormal basis.

Proof. By Lecture 5 (where triangularizability over $\mathbb{C}$ is sketched), there is *a* basis $v_1, \ldots, v_n$ in which T is upper-triangular, i.e. each subspace $U_k = \operatorname{span}(v_1, \ldots, v_k)$ is invariant, $TU_k \subseteq U_k$. Gram–Schmidt (Lecture 8) produces an orthonormal $e_1, \ldots, e_n$ with $\operatorname{span}(e_1, \ldots, e_k) = U_k$ for every k, so the same subspaces stay invariant; in this orthonormal basis T is upper-triangular. ■

Example 10.2 (Triangularizing in an orthonormal basis). $A = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ has the single eigenvalue 1 with eigenvector $(0, 1)$; taking the orthonormal basis $e_1 = (0, 1)$, $e_2 = (1, 0)$ gives $\mathcal{M}(A) = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, upper-triangular as Schur promises—but not diagonal, since A is not normal. ◁

2 The complex spectral theorem

> **Theorem 10.3 (Complex spectral theorem).** Suppose V is finite-dimensional over $\mathbb{C}$ and $T \in \mathcal{L}(V)$. The following are equivalent:
>
> 1. T is normal;
> 2. T has a diagonal matrix with respect to some orthonormal basis;
> 3. V has an orthonormal basis consisting of eigenvectors of T.

Proof. (2)$\Leftrightarrow$(3) is immediate, as the columns of a diagonal matrix in a basis say exactly that the basis vectors are eigenvectors. (2)$\Rightarrow$(1): a diagonal matrix and its conjugate transpose commute, so $T^\dagger T = TT^\dagger$.

(1)$\Rightarrow$(2): by Schur, take an orthonormal basis $e_1, \dots, e_n$ with $\mathcal{M}(T) = A$ upper-triangular. From the first column, $\|Te_1\|^2 = |a_{11}|^2$, while $\mathcal{M}(T^\dagger) = A^\dagger$ gives, from its first column (the conjugated first *row* of A), $\|T^\dagger e_1\|^2 = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2$. Normality forces $\|Te_1\| = \|T^\dagger e_1\|$ (Lecture 9), so $a_{12} = \cdots = a_{1n} = 0$. Hence $A = a_{11} \oplus A_1$. Comparing the two diagonal blocks of $A^\dagger A = AA^\dagger$ gives $A_1^\dagger A_1 = A_1 A_1^\dagger$, so the trailing upper-triangular block A_1 is normal. Induction on the dimension makes A_1 diagonal; therefore A is diagonal. ∎

In matrix language: A is normal iff $A = UDU^\dagger$ with U unitary and D diagonal. The columns of U are the orthonormal eigenvectors; conjugating by the unitary U (a change of orthonormal basis, Lecture 9) reveals the diagonal D.

$$\begin{tikzcd}
V \arrow[r, "T"] & V \\
V \arrow[u, "U"] \arrow[r, "D = \mathrm{diag}(\lambda_k)"'] & V \arrow[u, "U"']
\end{tikzcd}$$

Figure 1: The spectral theorem: a normal operator is the diagonal D seen through a *unitary* change of orthonormal basis U, so $T = UDU^\dagger$. Compare the non-orthogonal $A = PDP^{-1}$ of Lecture 6.

Example 10.4 (Unitary diagonalization of a normal operator). The normal operator $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ of Lecture 9 has eigenvalues $1 \pm i$ with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, i)$ and $\frac{1}{\sqrt{2}}(1, -i)$. With these as the columns of a unitary U,

$$U^\dagger A U = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}.$$

The operator is diagonal in an orthonormal basis, exactly as the theorem guarantees—no eigenvalue need be real, only normal. ◁

Example 10.5 (The unitary that diagonalizes). For that same A, the orthonormal eigenvectors are the columns of $U = \frac{1}{\sqrt{2}} \left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$, and one verifies $U^\dagger A U = \mathrm{diag}(1+i, 1-i)$. The columns of U *are* the eigenstates, and $U^\dagger$ rotates the standard basis into the eigenbasis. ◁

Example 10.6 (Diagonalizing a unitary). The rotation $R = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ is normal, hence unitarily diagonalizable: in the orthonormal eigenbasis $\frac{1}{\sqrt{2}}(1, \mp i)$ it becomes

$$\mathrm{diag}(e^{i\theta}, e^{-i\theta}),$$

a diagonal of unit-modulus phases. A real rotation and a pair of conjugate phases are the same operator in two bases. ◁

Example 10.7 (Circulant matrices and the Fourier basis). A circulant matrix—cyclically constant along diagonals, e.g. $\begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}$—is normal, and *every* circulant is diagonalized by the one orthonormal Fourier basis $f_k = \frac{1}{\sqrt{n}}(1, \omega^k, \omega^{2k}, \dots)$ with $\omega = e^{2\pi i/n}$. Its eigenvalues are the discrete Fourier transform of the first row. This is why the FFT diagonalizes convolution: a single unitary serves all circulants at once. ◁

Remark 10.8 (The spectrum is a complete invariant). Two normal operators are unitarily equivalent if and only if they have the same eigenvalues with the same multiplicities, since each is unitarily equivalent to the diagonal matrix of its eigenvalues. For normal operators the spectrum (with multiplicity) is therefore a complete invariant—unlike the non-normal case, where Jordan structure also matters.

Remark 10.9 (Bridge: commuting normal operators). A finite commuting family of normal operators on a finite-dimensional complex space can be simultaneously unitarily diagonalized. Diagonalize one operator. Every other operator B preserves each of its eigenspaces by commutation. Since their orthogonal direct sum is all of V, B is block diagonal in this decomposition, so $B^\dagger$ has the adjoint blocks; comparing the blocks in $B^\dagger B = BB^\dagger$ shows that every restriction of B is normal. Diagonalize the restrictions. Each remaining operator preserves the resulting common eigenspaces, so the same block argument iterates through the finite family and produces one orthonormal basis of joint eigenvectors. The core measurement application below uses the self-adjoint pair case.

Remark 10.10 (Why normality is needed). Normality is not optional. The shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not normal and has only a one-dimensional eigenspace, so no basis—let alone an orthonormal one—consists of its eigenvectors. The spectral theorem is precisely the statement that normal is the exact dividing line for unitary diagonalizability.

3 Self-adjoint operators and the real case

> **Theorem 10.11 (Spectral theorem for self-adjoint operators).** A self-adjoint operator on a finite-dimensional inner product space (real or complex) has an orthonormal basis of eigenvectors, with all eigenvalues real. Equivalently, a Hermitian matrix factors as $A = UDU^\dagger$ with U unitary and D real diagonal; a real symmetric matrix factors as $A = ODO^\mathsf{T}$ with O orthogonal.

Proof. A self-adjoint operator is normal, so over $\mathbb{C}$ the complex spectral theorem gives an orthonormal eigenbasis, and the eigenvalues are real by Lecture 9.

Over $\mathbb{R}$ the argument needs one extra step, because Theorem 10.1 as stated covers complex spaces. Let A be the real symmetric matrix of T in an orthonormal basis and regard it for a moment as a complex Hermitian matrix. Its characteristic polynomial is the same either way, and by Lecture 9 all of its roots are real; hence χ_T splits over $\mathbb{R}$. The proof of Theorem 10.1 uses nothing about $\mathbb{C}$ beyond the fact that χ_T splits—it triangularizes and then runs Gram–Schmidt—so it applies verbatim over $\mathbb{R}$ and yields a real orthonormal basis in which T is upper-triangular. A real symmetric upper-triangular matrix equals its transpose, hence is diagonal: an orthogonal eigenbasis with real entries. ■

For the finite-dimensional ideal-projective model this is the crucial statement: every self-adjoint observable has a complete orthonormal set of eigenstates with real eigenvalues. General observables with continuous spectrum need the projection-valued measures of Lecture 14. The real version is the "principal axes theorem"—a real symmetric matrix (inertia tensor, stress tensor, quadratic form) is diagonal in a suitable set of mutually perpendicular axes.

Example 10.12 (A self-adjoint operator diagonalized). The real symmetric matrix

$$A = \begin{pmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{pmatrix}$$

has real eigenvalues $27, 9, -15$ with orthonormal eigenvectors

$$\tfrac{1}{\sqrt{2}}(1,-1,0), \qquad \tfrac{1}{\sqrt{3}}(1,1,1), \qquad \tfrac{1}{\sqrt{6}}(1,1,-2).$$

Placing these as the columns of an orthogonal O gives $O^{\mathsf T}AO = \operatorname{diag}(27,9,-15)$—a complete orthonormal eigenbasis with real eigenvalues. ◁

Example 10.13 (A Hermitian 2×2). $A = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}$ is Hermitian, with real eigenvalues $1, 3$ and orthonormal eigenvectors $e_1 = \frac{1}{\sqrt{2}}(1, i)$, $e_2 = \frac{1}{\sqrt{2}}(1,-i)$. With $U = (e_1\, e_2)$, $U^\dagger AU = \operatorname{diag}(1,3)$: a two-state observable in its eigenbasis. ◁

Example 10.14 (A diagonalization from scratch). For the Hermitian $A = \begin{pmatrix} 3 & 1-i \\ 1+i & 2 \end{pmatrix}$ the characteristic equation is

$$(3-\lambda)(2-\lambda) - |1-i|^2 = \lambda^2 - 5\lambda + 4 = 0,$$

giving real eigenvalues $\lambda = 4, 1$. Solving $(A-\lambda I)v = 0$ yields the orthonormal eigenvectors $\frac{1}{\sqrt{3}}(1-i, 1)$ and $\frac{1}{\sqrt{3}}(1, -(1+i))$; with these as the columns of a unitary U, $U^\dagger AU = \operatorname{diag}(4,1)$. Every Hermitian operator reduces to a real diagonal this way. ◁

Example 10.15 (Principal axes of a quadratic form). The quadratic form $q(x,y) = 2x^2 + 2y^2 + 2xy$ has matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1,1)$, $\frac{1}{\sqrt{2}}(1,-1)$ and eigenvalues $3, 1$. In the rotated coordinates (x', y') along those axes, $q = 3x'^2 + y'^2$: a real symmetric matrix is diagonalized by a rotation, straightening a tilted ellipse into one aligned with the axes. ◁

Example 10.16 (A degenerate self-adjoint operator). $A = \operatorname{diag}(4,4,1)$ on $\mathbb{R}^3$ is self-adjoint with eigenvalue 4 (multiplicity 2) and 1. The 4-eigenspace is the whole xy-plane, so *any* orthonormal basis of that plane, together with e_3, diagonalizes A. Degeneracy leaves freedom in the eigenbasis—freedom fixed in physics by a further commuting observable. ◁

Example 10.17 (Normal modes from the real spectral theorem). The stiffness matrix $K = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ of two coupled masses is real symmetric, so the real spectral theorem furnishes an *orthogonal* eigenbasis: $\frac{1}{\sqrt{2}}(1,1)$ with $\omega^2 = 1$ and $\frac{1}{\sqrt{2}}(1,-1)$ with $\omega^2 = 3$. In these normal coordinates the coupled system $\ddot{x} = -Kx$ decouples into independent oscillators (Lecture 6)—the modes are perpendicular precisely because K is self-adjoint. ◁

Example 10.18 (Positive operators have nonnegative spectrum). $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ is self-adjoint with eigenvalues $3, 1 > 0$, hence *positive*: in eigenbasis coordinates $\langle v, Av\rangle = 3|c_1|^2 + |c_2|^2 \ge 0$. Positive operators are exactly the self-adjoint ones with nonnegative spectrum—the operator analogue of the nonnegative reals. Every positive operator has a unique *positive* square root; merely having a real matrix square root does not characterize positivity (for example, a real quarter-turn squares to $-I_2$). ◁

4 Spectral decomposition

Group the eigenvalues into the distinct values $\lambda_1, \ldots, \lambda_m$, and let P_k be the orthogonal projection onto the eigenspace $E(\lambda_k, T)$.

Theorem 10.19 (Spectral decomposition). A normal operator T on a finite-dimensional complex space (or a self-adjoint operator on a finite-dimensional real space) with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ satisfies

$$P_j P_k = \delta_{jk} P_k, \qquad P_1 + \cdots + P_m = I, \qquad T = \lambda_1 P_1 + \cdots + \lambda_m P_m,$$

with each P_k self-adjoint ($P_k^\dagger = P_k$). Consequently $f(T) = \sum_k f(\lambda_k) P_k$ for any function f on the spectrum.

Proof. The eigenspaces are mutually orthogonal (Lecture 9) and, by the spectral theorem, span V, so $V = E(\lambda_1) \oplus \cdots \oplus E(\lambda_m)$ orthogonally. The P_k are then orthogonal projections onto orthogonal summands, giving $P_j P_k = \delta_{jk} P_k$ and $\sum_k P_k = I$. Applying both sides of the last equation to an eigenvector shows $T = \sum_k \lambda_k P_k$, and orthogonal projections are self-adjoint. Applying f eigenvalue by eigenvalue gives $f(T) = \sum_k f(\lambda_k) P_k$. ■

In the same finite setting, with an orthonormal eigenbasis $\{|e_k\rangle\}$,

$$T = \sum_k \lambda_k |e_k\rangle\langle e_k|, \qquad I = \sum_k |e_k\rangle\langle e_k| :$$

the operator is a weighted sum of rank-one basis projectors, and the identity is their sum (the completeness relation of Lecture 8). When an eigenvalue is degenerate, its one physical ideal-projective outcome uses the higher-rank sum of all these basis projectors inside that eigenspace.

Example 10.20 (Spectral decomposition of σ_z). $\sigma_z = \mathrm{diag}(1, -1)$ has eigenprojections $P_+ = |0\rangle\langle 0| = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and $P_- = |1\rangle\langle 1| = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, with $P_+ + P_- = I$, $P_+ P_- = 0$, and $\sigma_z = (+1)P_+ + (-1)P_-$. Any function follows at once: $e^{i\theta\sigma_z} = e^{i\theta} P_+ + e^{-i\theta} P_- = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. ◁

Example 10.21 (Spectral decomposition of σ_x). σ_x has eigenvalues ± 1 with projectors $P_\pm = |\pm\rangle\langle\pm| = \frac{1}{2}\left(\begin{smallmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{smallmatrix}\right)$, so $\sigma_x = P_+ - P_-$ and $f(\sigma_x) = f(1)P_+ + f(-1)P_-$ for any f. Taking $f(\lambda) = e^{i\theta\lambda}$ recovers $e^{i\theta\sigma_x} = \cos\theta\, I + i \sin\theta\, \sigma_x$ from Lecture 6, now read straight off the spectral form. ◁

Example 10.22 (A function via the spectral form). For the Hermitian A of Example 10.13, with eigenprojections P_1, P_3 onto the eigenvalue-1 and eigenvalue-3 spaces, $\sqrt{A} = P_1 + \sqrt{3}\, P_3$ and $e^A = e\, P_1 + e^3 P_3$. Because A is positive, $\sqrt{A}$ is a genuine positive square root—the spectral calculus of Lecture 6 in orthonormal form. ◁

Example 10.23 (A positive square root, explicitly). The positive operator $A = \left(\begin{smallmatrix} 5 & 3 \\ 3 & 5 \end{smallmatrix}\right)$ has eigenvalues $8, 2$ with orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, \pm 1)$, so $P_\pm = \frac{1}{2}(I \pm \sigma_x)$ and

$$\sqrt{A} = \sqrt{8}\, P_+ + \sqrt{2}\, P_- = \frac{\sqrt{8} + \sqrt{2}}{2} I + \frac{\sqrt{8} - \sqrt{2}}{2} \sigma_x,$$

the unique positive operator squaring to A. ◁

Example 10.24 (Eigenprojections written out). For $A = \left(\begin{smallmatrix} 2 & i \\ -i & 2 \end{smallmatrix}\right)$ of Example 10.13, the eigenprojections are

$$P_1 = |e_1\rangle\langle e_1| = \tfrac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}, \qquad P_3 = |e_2\rangle\langle e_2| = \tfrac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix},$$

which satisfy $P_1 + P_3 = I$, $P_1 P_3 = 0$, and $A = 1 \cdot P_1 + 3 \cdot P_3$. ◁

Remark 10.25 (Projections by interpolation). The eigenprojections can be written without finding eigenvectors: $P_k = \prod_{j\neq k} \dfrac{A-\lambda_j I}{\lambda_k-\lambda_j}$, the Lagrange polynomial equal to 1 at λ_k and 0 at the other eigenvalues. For σ_z this gives $P_\pm = \frac{1}{2}(I \pm \sigma_z)$ at once.

Example 10.26 (Spectral decomposition of σ_y). σ_y has eigenvalues ±1 with eigenvectors $\frac{1}{\sqrt{2}}(1, \pm i)$, hence projectors $P_\pm = \frac{1}{2}\left(\begin{smallmatrix} 1 & \mp i \\ \pm i & 1 \end{smallmatrix}\right)$ and $\sigma_y = P_+ - P_-$. The three Pauli operators share the eigenvalues ±1, but projector families for different axes fail to commute—that failure, not mere difference of the families, makes the corresponding ideal observables incompatible. ◁

Example 10.27 (Reconstructing an operator). Conversely, spectral data determine the operator: from eigenvalues 2, 5 with projectors $P_2 = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $P_5 = \frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$,

$$A = 2P_2 + 5P_5 = \frac{1}{2}\begin{pmatrix} 7 & -3 \\ -3 & 7 \end{pmatrix}.$$

The operator and its spectral data are interchangeable. ◁

Example 10.28 (A projector resolution in $\mathbb{C}^3$). For $A = \operatorname{diag}(2, 2, 5)$ on $\mathbb{C}^3$ the eigenprojections are $P_2 = \operatorname{diag}(1, 1, 0)$ and $P_5 = \operatorname{diag}(0, 0, 1)$, with $P_2 + P_5 = I$, $P_2P_5 = 0$, and $A = 2P_2 + 5P_5$. The degenerate eigenvalue 2 carries a rank-2 projector, and $\operatorname{tr} P_2 = 2$ counts its multiplicity. ◁

Example 10.29 (Inverse from the spectrum). A normal $T = \sum_k \lambda_k P_k$ with every $\lambda_k \neq 0$ is invertible, with $T^{-1} = \sum_k \lambda_k^{-1} P_k$: indeed $\left(\sum_k \lambda_k P_k\right)\left(\sum_j \lambda_j^{-1} P_j\right) = \sum_k P_k = I$, using $P_j P_k = \delta_{jk} P_k$. No function of an operator asks more than the same function of its eigenvalues. ◁

Remark 10.30 (The functional calculus). The map $f \mapsto f(T)$ is an algebra homomorphism: $(f+g)(T) = f(T) + g(T)$, $(fg)(T) = f(T)g(T)$, and $\overline{f}(T) = f(T)^\dagger$. Because $f(T) = \sum_k f(\lambda_k) P_k$, each of these reduces to the corresponding fact about numbers applied eigenvalue by eigenvalue—the *functional calculus* of a normal operator.

Remark 10.31 (Trace and determinant from the spectrum). Because λ_k indexes distinct eigenvalues, $\operatorname{tr} T = \sum_k \lambda_k \operatorname{rank}(P_k)$ and $\det T = \prod_k \lambda_k^{\operatorname{rank}(P_k)}$, both basis-independent (Lecture 5). Thus $\operatorname{diag}(2, 2, 5)$ has trace $2 \cdot 2 + 5 = 9$ and determinant $2^2 \cdot 5 = 20$. Equivalently $\operatorname{rank}(P_k) = \operatorname{tr} P_k$, so the trace follows directly from the spectral decomposition.

Remark 10.32 (Extension: singular value decomposition). Even a non-normal T yields to a spectral idea: applying the theorem to the positive operator $T^\dagger T$ gives the *singular value decomposition* $T = U\Sigma V^\dagger$ with U, V unitary and Σ diagonal with nonnegative entries—the universal factorization behind low-rank approximation and the pseudoinverse.

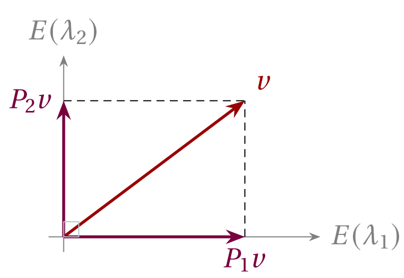

Figure 2: Spectral decomposition: the orthogonal eigenspaces split V into perpendicular summands, and $v = P_1 v + P_2 v + \cdots$ resolves every vector into its eigen-components. The observable acts on each piece by the scalar λ_k.

5 Measurement and what comes next

The finite spectral theorem supplies the projectors used by the ideal projective measurement postulate. For a normalized pure-state representative, the separate Born postulate says that an observable $A = \sum_k \lambda_k P_k$ yields outcome λ_k with probability $\|P_k|\psi\rangle\|^2$ (the Born rule), after which the state collapses to the normalized projection $P_k|\psi\rangle/\|P_k|\psi\rangle\|$ when that probability is nonzero (the Lueders update postulate). Because $\sum_k P_k = I$ and the P_k are orthogonal, these probabilities sum to one. Expectation values are $\langle A\rangle = \sum_k \lambda_k \|P_k|\psi\rangle\|^2$.

Example 10.33 (A measurement). Measuring σ_z on $|\psi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2|1\rangle)$: the outcome $+1$ occurs with probability $\|P_+|\psi\rangle\|^2 = \frac{1}{5}$ and -1 with probability $\frac{4}{5}$, collapsing $|\psi\rangle$ to $|0\rangle$ or $|1\rangle$ respectively. The expectation value is $\langle\sigma_z\rangle = (+1)\frac{1}{5} + (-1)\frac{4}{5} = -\frac{3}{5}$, the probability-weighted average of the eigenvalues. ◁

Example 10.34 (Measuring a degenerate observable). For $A = \mathrm{diag}(5, 5, 2)$ on $\mathbb{C}^3$ the eigenvalue 5 has a two-dimensional eigenspace with projection $P_5 = \mathrm{diag}(1, 1, 0)$. Measuring A on $|\psi\rangle = (a, b, c)$ returns 5 with probability $\|P_5\psi\|^2 = |a|^2 + |b|^2$, collapsing the state into the 5-eigenspace rather than to a single vector. Degenerate outcomes project onto whole eigenspaces. ◁

Example 10.35 (Time evolution via the spectral form). For a finite-dimensional, time-independent Hamiltonian $\hat{H} = \sum_n E_n P_n$,

$$e^{-i\hat{H}t/\hbar} = \sum_n e^{-iE_n t/\hbar} P_n,$$

so evolution acts on each energy eigenspace by a phase. A state $|\psi\rangle = \sum_n P_n|\psi\rangle$ becomes $\sum_n e^{-iE_n t/\hbar} P_n|\psi\rangle$, and the outcome probabilities $\|P_n|\psi\rangle\|^2$ are conserved—energy conservation, read off the spectral decomposition. ◁

Example 10.36 (Measuring in a rotated basis). Prepare $|0\rangle$ (a σ_z-eigenstate) and measure σ_x, whose eigenstates are $|\pm\rangle$. The amplitudes $\langle +, 0\rangle = \langle -, 0\rangle = \frac{1}{\sqrt{2}}$ give both outcomes ± 1 probability $\frac{1}{2}$: a state sharp in one observable is maximally uncertain in an incompatible one. ◁

Remark 10.37 (Extension: Schrödinger and Heisenberg pictures). One may evolve states by $e^{-i\hat{H}t/\hbar}$ or, equivalently, evolve observables by $A(t) = e^{i\hat{H}t/\hbar} A e^{-i\hat{H}t/\hbar}$. Conjugation by a unitary preserves the spectrum, so $A(t)$ has the same eigenvalues as A at all times: the possible outcomes never change, only their probabilities.

Example 10.38 (A three-outcome measurement). Measuring $A = \mathrm{diag}(1, 2, 3)$ on $|\psi\rangle = \frac{1}{\sqrt{3}}(1, 1, 1)$ gives each outcome $1, 2, 3$ with probability $\frac{1}{3}$, so

$$\langle A\rangle = \tfrac{1}{3}(1 + 2 + 3) = 2, \qquad \langle A^2\rangle = \tfrac{1}{3}(1 + 4 + 9) = \tfrac{14}{3},$$

and the variance is $\langle A^2\rangle - \langle A\rangle^2 = \frac{14}{3} - 4 = \frac{2}{3}$. The spectral data deliver every statistical moment of a measurement. ◁

Remark 10.39 (Repeated measurement). In the finite ideal-projective model, immediately repeating the same measurement returns the same outcome with certainty: the first Lueders update leaves the state in an eigenspace of A, on which A acts as a single eigenvalue. This is an implication of that physical update postulate, not of the spectral theorem alone.

Commuting finite-dimensional self-adjoint observables can be diagonalized *simultaneously* in one orthonormal basis, so a maximal set of commuting observables (a "complete set of commuting observables") labels states by joint eigenvalues—the quantum numbers of a system. Non-commuting observables share no eigenbasis, and the uncertainty relation of Lecture 9 measures how badly. The next lectures leave finite dimensions behind: dual spaces and tensors (Lectures 11–12), then Hilbert spaces and the spectral theorem for operators with continuous spectra (Lectures 13–14), where position and momentum finally find their home.

Example 10.40 (Extension: a complete set of commuting observables). On $\mathbb{C}^3$, $A = \operatorname{diag}(1,1,2)$ and $B = \operatorname{diag}(3,5,5)$ commute and share the standard orthonormal eigenbasis. Neither alone separates all three states—A cannot tell e_1 from e_2, B cannot tell e_2 from e_3—but the *joint* eigenvalues $(1,3)$, $(1,5)$, $(2,5)$ are distinct, so together A, B label the states completely, a finite model of a complete set of commuting observables. ◁

Example 10.41 (A shared eigenbasis). $A = \sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ commute, since $AB = BA = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, so they share the orthonormal eigenbasis $|\pm\rangle = \frac{1}{\sqrt{2}}(1, \pm 1)$: there $A = \operatorname{diag}(1,-1)$ and $B = \operatorname{diag}(3,1)$. Commuting observables diagonalize together even when neither is diagonal to begin with. ◁

Remark 10.42 (Extension: density matrices). A statistical mixture of states is a *density matrix* $\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$, a positive self-adjoint operator with $\operatorname{tr}\rho = 1$. By the spectral theorem ρ has an orthonormal eigenbasis with nonnegative eigenvalues summing to one—its own probability distribution over orthogonal pure states. The spectral theorem thus governs mixed states as well as pure ones.

Remark 10.43 (Beyond finite dimensions). The position operator has no eigenvectors in the usual sense: its spectrum is the *continuous* range of possible positions. The finite sum $A = \sum_k \lambda_k P_k$ then becomes an integral $\int \lambda \, dP(\lambda)$ against a projection-valued measure—the spectral theorem of Lecture 14, for which the finite-dimensional version proved here is the template.

Summary of main concepts

The finite spectral theorem says a normal operator on a finite-dimensional complex space has an orthonormal basis of eigenvectors—equivalently a diagonal matrix in an orthonormal basis, $T = UDU^\dagger$ with U unitary. Self-adjoint operators are the special case with real eigenvalues (and a real orthogonal eigenbasis when the space is real), the principal-axes theorem. Grouping eigenspaces gives the spectral decomposition $T = \sum_k \lambda_k P_k$ with orthogonal projections satisfying $P_j P_k = \delta_{jk} P_k$ and $\sum_k P_k = I$, whence $f(T) = \sum_k f(\lambda_k) P_k$. In the finite quantum model this supplies real spectral values and eigenspace projectors. The separate Born and Lueders postulates give ideal-projective probabilities and conditional updates, and commuting observables diagonalize together. Continuous spectra and generalized measurements are deferred to Lecture 14.

- **Schur's theorem**—every finite-dimensional complex operator is upper-triangular in some orthonormal basis.
- **Complex spectral theorem**—in finite dimension, normal $\Longleftrightarrow$ orthonormal eigenbasis $\Longleftrightarrow$ unitarily diagonalizable ($T = UDU^\dagger$).
- **Real spectral theorem**—in finite dimension, self-adjoint gives a real orthonormal eigenbasis (principal axes) and real eigenvalues.
- **Spectral decomposition**—the finite sum $T = \sum_k \lambda_k P_k$ has $P_j P_k = \delta_{jk} P_k$, $\sum_k P_k = I$; $f(T) = \sum_k f(\lambda_k) P_k$.
- **Finite projective model**—the theorem supplies projectors; Born probabilities and the Lueders update are separate postulates; commuting observables diagonalize together.

Exercises for this lecture are in the companion sheet `exercises-10.pdf`.